

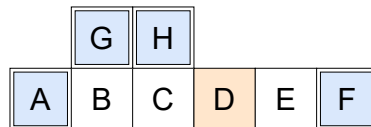
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QUIZ 3**DURATION: 30 Minutes****WINTER SEMESTER, 2020-2021****FULL MARKS: 30****CSE 4711: Artificial Intelligence**There are **2 (two)** questions. Answer **both** of them.

Figures in the right margin indicate marks for each question.

The square brackets at the start of each question denote the corresponding CO and PO.

1. Consider that an agent is in cell *D* of the Gridworld shown in Figure 1. The only action available in cells *A*, *F*, *G*, and *H* is `Exit` with rewards 6, 3, 0, and 0, respectively. When the agent exits from a cell, it gets the reward for that cell as specified. From the other cells, the agent can only take the action `Left` or `Right`, which results in the agent moving to the immediate left or right cell, respectively.

**Figure 1:** Gridworld for Question 1

If the agent is in cell *B* or *C*, and it takes a `Left` or `Right` action, it might fail sometimes. The agent will move in the desired direction with probability p , and it will fail and move up with probability $1 - p$. If the agent is in any other cell (*A*, *D*, *E*, *F*, *G*, or *H*), the action will always be successful.

Assume that there is no living reward/penalty and the discount factor is γ .

- a) [CO2, PO4] Model the problem as an MDP. Show the set of states, set of actions, start state, and terminal state(s). 8

Solution:

- States, $S = \{A, B, C, D, E, F, G, H\}$
- Actions, $A = \{\text{Left}, \text{Right}, \text{Exit}\}$
- Start State: *D*
- Terminal States: $\{A, F, G, H\}$

Rubric:

- 0.5 points for each correct value

- b) [CO1, PO1] For each of the following policies, determine the value of each non-terminal state. Assume $\gamma \in [0, 1]$. 6 × 2

i. $\pi_{\text{Right}}(S) = \begin{cases} \text{Exit}, & \text{if } S \text{ is a Terminal State} \\ \text{Right}, & \text{otherwise} \end{cases}$

Solution:

- $B \rightarrow 3\gamma^4 p^2$

- $C \rightarrow 3\gamma^3 p$

- $D \rightarrow 3\gamma^2$

- $E \rightarrow 3\gamma$

Rubric:

- 0.5 points for each correct integer
- 0.5 points for each correct discount
- 0.5 points for each correct probability

ii. $\pi_{\text{Left}}(S) = \begin{cases} \text{Exit}, & \text{if } S \text{ is a Terminal State} \\ \text{Left}, & \text{otherwise} \end{cases}$

Solution:

- $B \rightarrow 6\gamma p$

- $C \rightarrow 6\gamma^2 p^2$

- $D \rightarrow 6\gamma^3 p^2$

- $E \rightarrow 6\gamma^4 p^2$

Rubric:

- 0.5 points for each correct integer
- 0.5 points for each correct discount
- 0.5 points for each correct probability

c) [CO3, PO2, PO3] What values of p will ensure that going Right from the start state is the optimal action? Provide a range in terms of γ .

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Solution: For going right to be optimal, we need $V^{\pi_{\text{Right}}}$ of the start state to be greater or equal to $V^{\pi_{\text{Left}}}$. We get:

$$V^{\pi_{\text{Right}}}(D) \geq V^{\pi_{\text{Left}}}(D)$$

$$3\gamma^2 \geq 6\gamma^3 p^2$$

$$6\gamma^3 p^2 \leq 3\gamma^2$$

$$p^2 \leq \frac{3}{6\gamma}$$

$$p \leq \frac{1}{\sqrt{2\gamma}}$$

We get: $p \in \left[0, \frac{1}{\sqrt{2\gamma}}\right]$

Rubric:

- For defining the correct relation between the value of the start state: 3 points
- For the correct lower range: 1 point
- For the correct upper range: 1 point

2. [CO1, CO2, CO3, PO1, PO2, PO3, PO4] Write everything that you know about MDPs that was not asked in this quiz!

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Solution: Rubric:

- Any correct information provided here that is not related to the questions given are accepted.